

Probability & Stochastic Processes

Theorem Cheat Sheet

Every Named Theorem, Definition, Worked Example & When/How to Use It

Chapters 4 (§4.2–4.3, 4.6–4.7) · 5 (§5.7–5.9) · 6 (§6.2–6.3) · 13 (§13.6–13.11)

Plus every problem from Exam 1 and Exam 2, fully solved

How to read this sheet. Every entry gives the exact textbook number/name, the formula, a short *worked numeric example*, and a note on *when you'd reach for it on an exam and how to apply it*. Entries the professor personally flagged (Theorem 13.10, 13.11, 13.12, Example 13.22, and the bivariate Gaussian PDF) are marked ★ in red double-ruled boxes — know these cold. Part I at the end solves every Exam 1 and Exam 2 problem, tagged with the exact theorems used above. This is a companion to the full *Final_Exam_Guide.pdf*, which has complete proofs as well.

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1 Chapter 4 — CDF, PDF, Gaussian RVs, Delta Functions

1.1 CDF and PDF Foundations (§4.2–4.3)

Definition 4.1 — Cumulative Distribution Function

$$F_X(x) = P[X \leq x].$$

Example: Pointer spun on a 1-meter circle, $X \in [0, 1)$ uniform: $F_X(x) = 0$ for $x < 0$, $F_X(x) = x$ for $0 \leq x < 1$, $F_X(x) = 1$ for $x \geq 1$.

Use it when / how: The universal probability model — works for discrete, continuous, or mixed X . Reach for this first whenever a problem defines a random variable in terms of “the probability that X is at most x ,” or when you need $P[a < X \leq b]$.

Theorem 4.1 — Basic CDF Properties

$$F_X(-\infty) = 0, F_X(\infty) = 1, P[x_1 < X \leq x_2] = F_X(x_2) - F_X(x_1).$$

Example: For the pointer above, $P[1/4 < X \leq 3/4] = F_X(3/4) - F_X(1/4) = 3/4 - 1/4 = 1/2$.

Use it when / how: Use property (c) any time you're asked for an interval probability and you already have (or can derive) F_X . Use (a)/(b) as a sanity check on any CDF you derive — if it doesn't go from 0 to 1, you made an error.

Definition 4.3 — Probability Density Function

$$f_X(x) = dF_X(x)/dx.$$

Example: Differentiating $F_X(x) = x$ on $(0, 1)$ gives $f_X(x) = 1$ on $(0, 1)$, 0 elsewhere — the uniform PDF.

Use it when / how: Differentiate a CDF to get the PDF; conversely integrate a PDF to get the CDF (Theorem 4.2). This is the two-way bridge you use constantly in Chapter 6 problems.

Theorem 4.2 — PDF Properties

$$f_X(x) \geq 0; F_X(x) = \int_{-\infty}^x f_X(u) du; \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Example: Given $f_X(x) = c(1-x^2)$ on $[-1, 1]$ (Exam 1, Problem 4), normalization forces $c \int_{-1}^1 (1-x^2) dx = \frac{4}{3}c = 1 \Rightarrow c = \frac{3}{4}$.

Use it when / how: The normalization property ($\int f_X = 1$) is exactly how you solve “find the constant c/A ” problems — set the total integral (or sum, for PMFs) equal to 1 and solve.

Theorem 4.3 — Interval Probability from the PDF

$$P[x_1 < X \leq x_2] = \int_{x_1}^{x_2} f_X(x) dx.$$

Example: With $f_X(x) = 1$ on $(0, 1)$: $P[1/4 < X < 3/4] = \int_{1/4}^{3/4} 1 dx = 1/2$.

Use it when / how: Whenever you have f_X (not yet F_X) and need a probability over a range, integrate directly instead of first finding the full CDF — faster when you only need one interval.

Definition 4.4 — Expected Value

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Example: For $f_X(x) = 1$ on $(0, 1)$: $E[X] = \int_0^1 x dx = 1/2$ meter — the average pointer position.

Use it when / how: The “average value” of X . Always pair with Theorem 4.5(c) to get variance — never integrate $(x - \mu)^2 f_X(x)$ directly if you can avoid it.

Theorem 4.4 — Law of the Unconscious Statistician (LOTUS)

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

Example: For $f_X(x) = 1$ on $(0, 1)$: $E[X^2] = \int_0^1 x^2 dx = 1/3$, obtained directly without ever finding the PDF of X^2 .

Use it when / how: **The single biggest time-saver in the course.** Use this whenever a question asks only for $E[g(X)]$ (e.g. $E[X^2]$, $E[e^X]$) — you never need to derive the PDF of $g(X)$ first. Only derive the full PDF of $W = g(X)$ (Chapter 6 machinery) if the question asks for the whole distribution of W , not just its mean.

Theorem 4.5 — Linearity and Variance Identities

$$E[X - \mu_X] = 0; \quad E[aX + b] = aE[X] + b; \quad \text{Var}[X] = E[X^2] - \mu_X^2; \quad \text{Var}[aX + b] = a^2 \text{Var}[X].$$

Example: Continuing the pointer: $E[X] = 1/2$, $E[X^2] = 1/3 \Rightarrow \text{Var}[X] = 1/3 - 1/4 = 1/12$.

Use it when / how: (c) is how *every* variance computation is actually done on this exam: find $E[X^2]$ and $E[X]$ separately, then subtract the square. (b)/(d) let you rescale a random variable's mean/variance instantly without re-deriving anything.

1.2 Standard Continuous Families — Mean and Variance for Each (§4.5)

Definition 4.5 & Theorem 4.6 — Uniform(a, b): PDF, CDF, $E[\cdot]$, $\text{Var}[\cdot]$

$$f_X(x) = \frac{1}{b-a} \text{ on } (a, b); \quad F_X(x) = \frac{x-a}{b-a} \text{ on } [a, b]; \quad E[X] = \frac{a+b}{2}, \quad \text{Var}[X] = \frac{(b-a)^2}{12}.$$

Example: Modem phase $\Theta \sim \text{Uniform}(0, 2\pi)$: $E[\Theta] = \pi$ rad, $\text{Var}[\Theta] = (2\pi)^2/12 = \pi^2/3$ — read straight off the formulas, no integration needed.

Use it when / how: Recognize “uniform on (a, b) ” immediately and pull $E[\cdot]$, $\text{Var}[\cdot]$ from the boxed formulas — never re-derive them by integrating $x/(b-a)$ from scratch.

Definition 4.6 & Theorem 4.8 — Exponential(λ): PDF, CDF, $E[\cdot]$, $\text{Var}[\cdot]$

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0; \quad F_X(x) = 1 - e^{-\lambda x}; \quad E[X] = \frac{1}{\lambda}, \quad \text{Var}[X] = \frac{1}{\lambda^2}.$$

Example: Call duration T with $F_T(t) = 1 - e^{-t/3} \Rightarrow \lambda = 1/3 \Rightarrow E[T] = 3$ min, $\text{Var}[T] = 9$ min², $\sigma_T = 3$ min.

Use it when / how: Recognize $F_X(x) = 1 - e^{-\lambda x}$ (or $f_X = \lambda e^{-\lambda x}$) instantly as exponential and read off the rate λ — this is the standard “waiting time” / “arrival gap” model.

Definition 4.7 & Theorem 4.10 — Erlang(n, λ): PDF, $E[\cdot]$, $\text{Var}[\cdot]$

$$f_X(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!}, \quad x \geq 0; \quad E[X] = \frac{n}{\lambda}, \quad \text{Var}[X] = \frac{n}{\lambda^2}.$$

Example: Sum of $n = 4$ iid Exponential($\lambda = 2$) waiting times is Erlang(4, 2): $E[X] = 4/2 = 2$, $\text{Var}[X] = 4/4 = 1$ — Erlang(1, λ) reduces to plain Exponential(λ).

Use it when / how: Recognize a sum of n iid exponential(λ) random variables (e.g. “time until the n th arrival”) as Erlang(n, λ) and read $E[\cdot]$, $\text{Var}[\cdot]$ directly — no convolution needed.

Theorems 4.7, 4.9, 4.11 — Bridges to Discrete Families

If $X \sim \text{Uniform}(a, b)$ (integer a, b), $K = \lceil X \rceil$ is discrete-uniform($a+1, b$) (Thm 4.7). If $X \sim \text{Exponential}(\lambda)$, $K = \lceil X \rceil$ is geometric(p) with $p = 1 - e^{-\lambda}$ (Thm 4.9). If $X \sim \text{Erlang}(n, \lambda)$, $F_X(x) = P[\text{Poisson}(\lambda x) \geq n]$ (Thm 4.11).

Example: $T \sim \text{Exponential}(1/3)$ (minutes): rounding *up* to the next full minute, $K = \lceil T \rceil$ is geometric(p) with $p = 1 - e^{-1/3} \approx 0.284$, so $E[K] = 1/p \approx 3.53$ minutes billed — more than $E[T] = 3$, since rounding up always costs extra.

Use it when / how: Use these whenever a continuous waiting-time model gets *rounded/discretized* (e.g. billing by the minute) or whenever you need to relate an Erlang CDF to a Poisson probability — they are the formal link between the continuous-time family (Ch. 4) and the discrete-time/counting family (Ch. 3, and later the Poisson process in Ch. 13).

1.3 Gaussian Random Variables (§4.6)

Definition 4.8 — Gaussian Random Variable

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \text{ written } X \sim N[\mu, \sigma^2].$$

Example: A test score is $X \sim N[61, 100]$ (i.e. $\mu = 61$, $\sigma = 10$).

Use it when / how: Recognize this bell-curve form immediately whenever a problem says “Gaussian,” “normal,” or gives you this exact exponential form (including inside a joint PDF — see §5.9 below).

Theorem 4.12 — Gaussian Mean and Variance

$$E[X] = \mu, \quad \text{Var}[X] = \sigma^2.$$

Example: $X \sim N[61, 100]$: read off instantly $E[X] = 61$, $\text{Var}[X] = 100$ ($\sigma = 10$).

Use it when / how: Read the parameters straight off the notation $N[\mu, \sigma^2]$ — no integration needed.

Theorem 4.13 — Linear Transform of a Gaussian

$$\text{If } X \sim N[\mu, \sigma^2], \text{ then } Y = aX + b \sim N[a\mu + b, a^2\sigma^2].$$

Example: If $Z \sim N[0, 1]$ and $X = 10Z + 61$, then $X \sim N[61, 100]$ — the standardizing transform run in reverse.

Use it when / how: Use whenever a problem defines a new r.v. as an affine function of a Gaussian (e.g. rescaled voltage, shifted temperature) — you get the new distribution instantly without re-deriving a PDF.

Definitions 4.9 & 4.10 — Standard Normal Z and Its CDF Φ

$$Z \text{ is the Gaussian}(0, 1) \text{ r.v.}; \quad \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

Example: By Theorem 4.13, standardizing $X \sim N[\mu, \sigma^2]$ via $Z = (X - \mu)/\sigma$ always produces exactly this $Z \sim N[0, 1]$ — which is why a single table of Φ values covers every Gaussian random variable.

Use it when / how: Every Gaussian probability calculation bottoms out in a table lookup of $\Phi(z)$ for this one specific Z — Theorem 4.14 (next) is simply the recipe for getting there from any $N[\mu, \sigma^2]$.

Theorem 4.14 — Gaussian CDF via Φ

$$F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right), \quad P[a < X \leq b] = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

Example: $X \sim N[61, 100]$: $P[51 < X < 71] = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 = 2(0.8413) - 1 = 0.683$.

Use it when / how: **The standard tool for every “find the probability” Gaussian question.** Standardize to $z = (x - \mu)/\sigma$, then look up $\Phi(z)$ in the table.

Theorem 4.15 — Symmetry of Φ

$$\Phi(-z) = 1 - \Phi(z).$$

Example: $P[X < 46] = \Phi\left(\frac{46-61}{10}\right) = \Phi(-1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.067$.

Use it when / how: Use whenever your z -score comes out negative but your table only lists Φ for positive z .

Definition 4.11 — Q -function (complementary CDF)

$$Q(z) = 1 - \Phi(z) = P[Z > z].$$

Example: Exam 1, Problem 5: bit-error probability with SNR= 4 is $Q(2) = 1 - \Phi(2) = 1 - 0.97725 = 0.02275$.

Use it when / how: **Always the right tool for “rare event” / tail / bit-error-rate questions** (e.g. $P[\text{noise exceeds threshold}]$) — these are phrased in terms of Q , not Φ , because Q resolves the small-tail-probability precision that Φ loses near 1.

1.4 Delta Functions & Mixed Random Variables (§4.7)

Definition 4.12 — Unit Impulse (Delta) Function

$\delta(x) = \lim_{\epsilon \rightarrow 0} d_\epsilon(x)$, with $\int_{-\infty}^{\infty} \delta(x) dx = 1$.

Example: $Y \in \{1, 2, 3\}$ each with probability $1/3$ has PDF $f_Y(y) = \frac{1}{3}\delta(y-1) + \frac{1}{3}\delta(y-2) + \frac{1}{3}\delta(y-3)$.

Use it when / how: Whenever a random variable has a point mass (discrete part) embedded in an otherwise continuous PDF, that point mass is represented as $p\delta(x-x_0)$ in $f_X(x)$.

Theorem 4.16 — Sifting Property

$$\int_{-\infty}^{\infty} g(x) \delta(x-x_0) dx = g(x_0).$$

Example: $E[Y] = \int y \left[\frac{1}{3}\delta(y-1) + \frac{1}{3}\delta(y-2) + \frac{1}{3}\delta(y-3) \right] dy = \frac{1}{3}(1) + \frac{1}{3}(2) + \frac{1}{3}(3) = 2$.

Use it when / how: Use this to evaluate $E[X]$, $E[g(X)]$, or any integral once impulses appear in the PDF — the impulse just “picks out” the integrand’s value at x_0 , turning the integral into plain evaluation.

Theorem 4.17 — Step/Delta Relationship

$$\int_{-\infty}^x \delta(v) dv = u(x), \text{ equivalently } \delta(x) = du(x)/dx.$$

Example: The CDF jump of $1/3$ at $y=1$ (from the example above) corresponds to the impulse term $\frac{1}{3}\delta(y-1)$ in $f_Y(y)$.

Use it when / how: Use to convert between a discrete RV’s PMF-based CDF (a staircase built from $u(x-x_i)$) and its impulse-based PDF ($\sum P_X(x_i)\delta(x-x_i)$), and vice versa.

Theorem 4.18 — Four Equivalent Ways to State “a Point Has Probability q ”

For any random variable X , these are equivalent: (a) $P[X=x_0] = q$; (b) $P_X(x_0) = q$; (c) $F_X(x_0^+) - F_X(x_0^-) = q$; (d) $f_X(x_0) = q\delta(0)$.

Example: Exam 1, Problem 1: the impulse $\frac{1}{8}\delta(x+2)$ in f_X (form (d)) is exactly the same fact as the CDF jump $F_X(-2) - F_X(-2^-) = \frac{1}{8}$ (form (c)) and $P[X=-2] = \frac{1}{8}$ (form (a)).

Use it when / how: Use whenever a problem switches between describing a point mass as “probability q at x_0 ,” “a jump of size q in F_X ,” or “an impulse $q\delta(x-x_0)$ in f_X ” — they are the same information in three notations; pick whichever is easiest for the step you’re on.

Definition 4.14 — Mixed Random Variable

X is mixed iff $f_X(x)$ contains *both* impulses and nonzero finite (continuous) values.

Example: Phone call duration: $1/3$ chance the call never connects ($Y=0$), else uniform(0, 3) minutes: $f_Y(y) = \frac{1}{3}\delta(y) + \frac{2}{9}$ on $[0, 3)$, giving $E[Y] = 1$ minute.

Use it when / how: Recognize a mixed RV whenever a problem describes “with probability p nothing happens (point mass), otherwise X is continuous on some range” — write f_X as (impulse term) + (continuous term), find the impulse weight from the point-mass probability, and the continuous piece from the conditional continuous density times its probability.

2 Chapter 5 — Multiple Random Variables

2.1 Marginal PMF/PDF — Foundational (§5.3, §5.5)

Marginal PMF and Marginal PDF

$$P_X(x) = \sum_{y \in S_Y} P_{X,Y}(x, y); \quad f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \text{ (and symmetrically for } P_Y(y), f_Y(y)\text{)}.$$

Example: $f_{X,Y}(x, y) = \frac{x+y}{3}$ on $0 \leq x \leq 1, 0 \leq y \leq 2$ (Exam 1, Problem 2): $f_X(x) = \int_0^2 \frac{x+y}{3} dy = \frac{2x+2}{3}$ on $0 \leq x \leq 1$.

Use it when / how: The mandatory first step whenever a problem gives you a joint model but asks about only one variable ($E[X]$, $\text{Var}[X]$, $P[X \in A]$, etc.) — sum out (discrete) or integrate out (continuous) the *other* variable, then treat the result as an ordinary single-variable PMF/PDF and apply the Chapter 3/4 tools you already know.

2.2 Expected Value of a Function of Two RVs (§5.7)

Theorem 5.9 — $E[g(X, Y)]$

Discrete: $\sum_x \sum_y g(x, y) P_{X,Y}(x, y)$. Continuous: $\iint g(x, y) f_{X,Y}(x, y) dx dy$.

Example: Transmission time $T = 8L/(1000R)$ with independent speed R and length L : $E[T] = 1.652$ seconds (Example 5.15) — note this is *not* equal to $g(E[R], E[L]) = 1.475$ seconds.

Use it when / how: The two-variable version of LOTUS (Theorem 4.4) — use whenever you need $E[g(X, Y)]$ for *any* function g (not just $X + Y$ or XY) and you have the joint model.

Theorem 5.10 — Linearity of Expectation (General Case)

$$E[a_1 g_1(X, Y) + \cdots + a_n g_n(X, Y)] = a_1 E[g_1(X, Y)] + \cdots + a_n E[g_n(X, Y)].$$

Example: $E[3X^2 - 2XY + 5] = 3E[X^2] - 2E[XY] + 5$ — each piece computed separately via Theorem 5.9, then recombined with its coefficient.

Use it when / how: Use whenever the quantity you need is a *linear combination* of several functions of X, Y (not just one g) — split it into pieces, apply Theorem 5.9 to each piece independently, and recombine. Theorem 5.11 below is the special case $g_1 = X, g_2 = Y, a_1 = a_2 = 1$.

Theorem 5.11 — Expected Value of a Sum

$$E[X + Y] = E[X] + E[Y].$$

Example: Exam 2, Problem 1 ($f_{X,Y} = 4xy$ on the unit square): $E[X] = E[Y] = 2/3 \Rightarrow E[X + Y] = 4/3$.

Use it when / how: Always true, even if X, Y are dependent, and needs only the marginals. Use this first before reaching for the joint PDF — it's very often all you need for an " $E[X + Y]$ " question.

Theorem 5.12 — Variance of a Sum

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y].$$

Example: Same problem: $\text{Var}[X] = \text{Var}[Y] = 1/18, \text{Cov}[X, Y] = 0 \Rightarrow \text{Var}[X + Y] = 1/18 + 1/18 + 0 = 1/9$.

Use it when / how: Unlike $E[X + Y]$, this *needs* the joint model (through the covariance term) unless you already know X, Y are independent (then $\text{Cov}[X, Y] = 0$ and it reduces to just $\text{Var}[X] + \text{Var}[Y]$). This contrast is a favorite exam trap — know it cold.

2.3 Covariance, Correlation, Independence (§5.8)

Definition 5.5 — Covariance

$$\text{Cov}[X, Y] = E[(X - \mu_X)(Y - \mu_Y)].$$

Example: For $f_{X,Y} = 4xy$ on the unit square: $E[XY] = 4/9$, $E[X]E[Y] = (2/3)^2 = 4/9 \Rightarrow \text{Cov}[X, Y] = 0$.

Use it when / how: Never compute this by direct integration if you can avoid it — use Theorem 5.16(a) instead ($\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$), which is how every real exam solution does it.

Definition 5.6 — Correlation Coefficient

$$\rho_{X,Y} = \text{Cov}[X, Y] / (\sigma_X \sigma_Y).$$

Example: Exam 2, Problem 2 ($f_{X,Y} = ce^{-(2x^2-4xy+4y^2)}$): matching to the bivariate Gaussian form gives $\rho_{X,Y} = 1/\sqrt{2}$.

Use it when / how: The unit-free measure of linear relationship. Compute $\text{Cov}[X, Y]$, σ_X , σ_Y separately, then divide. Always check your answer satisfies $-1 \leq \rho_{X,Y} \leq 1$ (Theorem 5.14) — if not, you have an arithmetic error.

Theorem 5.13 — ρ and Cov Under Linear Rescaling

$$\text{If } \hat{X} = aX + b, \hat{Y} = cY + d: \quad \rho_{\hat{X},\hat{Y}} = \rho_{X,Y} \text{ (unchanged!)}, \quad \text{Cov}[\hat{X}, \hat{Y}] = ac \text{Cov}[X, Y].$$

Example: Height/weight in cm/kg vs. inches/lbs: Cov changes by the unit-conversion factors, but $\rho_{X,Y}$ (e.g. ≈ 0.7) is exactly the same number in either unit system.

Use it when / how: Use whenever a problem changes units or rescales one or both variables — $\rho_{X,Y}$ is *invariant* (that's the whole point of normalizing by $\sigma_X \sigma_Y$), while $\text{Cov}[X, Y]$ scales by the product ac of the two scale factors.

Theorem 5.14 — Correlation Coefficient Is Bounded

$$-1 \leq \rho_{X,Y} \leq 1.$$

Example: $\rho_{X,Y} = 1/\sqrt{2} \approx 0.707$ from the example above satisfies $|0.707| \leq 1$ — passes the sanity check.

Use it when / how: Use as an automatic sanity check on every ρ you compute.

Theorem 5.15 — Correlation of a Perfect Linear Relationship

$$\text{If } Y = aX + b: \quad \rho_{X,Y} = -1 \text{ } (a < 0), \quad 0 \text{ } (a = 0), \quad 1 \text{ } (a > 0).$$

Example: If gain Y (dB) = $-1 \times \text{attenuation } X$ (dB) + b , then instantly $\rho_{X,Y} = -1$.

Use it when / how: Use to instantly answer “what is $\rho_{X,Y}$ ” questions where Y is given as an exact linear function of X — no computation needed.

Definition 5.7 — Correlation

$$r_{X,Y} = E[XY].$$

Example: Exam 2, Problem 5 (joint PMF $P_{X,Y}(x, y) = 1/(4x)$ for $1 \leq y \leq x \leq 4$): $r_{X,Y} = E[XY] = 5$.

Use it when / how: Compute this first (via Theorem 5.9 with $g(x, y) = xy$) — it feeds directly into Theorem 5.16(a) for covariance.

Theorem 5.16 — The Master Covariance/Variance Identities

$$(a) \text{Cov}[X, Y] = r_{X,Y} - \mu_X \mu_Y. \quad (b) \text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y]. \quad (c) \text{If } X = Y: \text{Cov}[X, Y] = \text{Var}[X], \quad r_{X,Y} = E[X^2].$$

Example: Continuing Exam 2 Problem 5: $r_{X,Y} = 5$, $E[X] = 5/2$, $E[Y] = 7/4 \Rightarrow \text{Cov}[X, Y] = 5 - (5/2)(7/4) = 10/16$.

Use it when / how: **This is the workhorse of every Chapter 5 exam problem.** Standard solution path: find $E[X]$, $E[Y]$, $E[XY]$ (via Theorem 5.9) $\rightarrow$ get $\text{Cov}[X, Y]$ via (a) $\rightarrow$ get $\rho_{X,Y}$ by dividing by $\sigma_X \sigma_Y \rightarrow$ get $\text{Var}[X + Y]$ via (b) if asked.

Definition 5.8/5.9 — Orthogonal vs. Uncorrelated

$$\text{Orthogonal: } r_{X,Y} = 0. \quad \text{Uncorrelated: } \text{Cov}[X, Y] = 0.$$

Example: Exam 2, Problem 4 ($f_{X,Y} = 1/2$ on $-1 \leq x \leq y \leq 1$): $r_{X,Y} = E[XY] = 0$, so X, Y are orthogonal; since also $E[X] = 0$ here, they're uncorrelated too.

Use it when / how: Watch the exact wording of a question — “orthogonal” asks about $E[XY]$, “uncorrelated” asks about $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$. They coincide only when $E[X] = 0$ or $E[Y] = 0$.

Theorem 5.17 — Independence $\Rightarrow$ Uncorrelated (Not Conversely)

Independent $X, Y \Rightarrow E[g(X)h(Y)] = E[g(X)]E[h(Y)]$, $r_{X,Y} = E[X]E[Y]$, $\text{Cov}[X, Y] = \rho_{X,Y} = 0$, $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$.

Example: Signal $X \sim N[0, \sigma_X^2]$ and independent noise $Z \sim N[0, \sigma_Z^2]$: $\text{Cov}[X, Z] = 0$ and $\text{Var}[X + Z] = \sigma_X^2 + \sigma_Z^2$ follow immediately from independence.

Use it when / how: Use the forward direction freely once independence is given/shown. **Never use the converse** — uncorrelated does **NOT** imply independent — unless the pair is bivariate Gaussian (Theorem 5.20), a very common exam trap otherwise.

2.4 Bivariate Gaussian Random Variables (§5.9)

★ Definition 5.10 — The Bivariate Gaussian PDF ★ [PROFESSOR-FLAGGED]

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho^2)} \right].$$

Example: Exam 2, Problem 2: matching $ce^{-(2x^2-4xy+4y^2)}$ term-by-term to this exponent gives $\sigma_X = 1/\sqrt{2}$, $\sigma_Y = 1/2$, $\rho = 1/\sqrt{2}$, and then $c = 1/(2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}) = 2/\pi$.

Use it when / how: **Memorize this exact formula.** When you're handed a joint PDF of the form $ce^{-(\text{quadratic in } x, y)}$, match the quadratic's coefficients term-by-term against the exponent above to read off σ_X, σ_Y, ρ (and hence c). It has 5 parameters: $\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho_{X,Y}$.

Theorem 5.18 — Marginals of a Bivariate Gaussian

If X, Y are bivariate Gaussian, $X \sim N[\mu_X, \sigma_X^2]$ and $Y \sim N[\mu_Y, \sigma_Y^2]$.

Example: From Exam 2 Problem 2: the marginal of X alone is $N[0, 1/2]$, regardless of ρ .

Use it when / how: Use whenever you need just the marginal distribution of X or Y from a bivariate Gaussian joint PDF — ρ drops out entirely for a single-variable marginal.

Theorem 5.19 — ρ Really Is the Correlation Coefficient

The parameter ρ in Definition 5.10 equals $\rho_{X,Y}$ as defined in Definition 5.6.

Example: In Exam 2 Problem 2, once you match coefficients and get $\rho = 1/\sqrt{2}$, that *is* $\text{Cov}[X, Y]/(\sigma_X\sigma_Y)$ — no separate computation needed.

Use it when / how: Justifies reading ρ directly off the bivariate Gaussian formula instead of computing $\text{Cov}[X, Y]/(\sigma_X\sigma_Y)$ from scratch.

★ Theorem 5.20 — Uncorrelated $\Leftrightarrow$ Independent (Bivariate Gaussian Only) ★ [PROFESSOR-FLAGGED]

Bivariate Gaussian X, Y : uncorrelated $\Leftrightarrow$ independent.

Example: Exam 2, Problem 2: since $\rho = 1/\sqrt{2} \neq 0$, X, Y are **dependent** — this theorem is exactly why $\rho \neq 0$ alone is enough to conclude dependence for a bivariate Gaussian pair.

Use it when / how: The one and only exception to Theorem 5.17's one-way arrow. Use this whenever the problem states or implies X, Y are jointly Gaussian — then (and only then) $\rho = 0$ lets you conclude independence.

Theorem 5.21 — Linear Combinations of Jointly Gaussian Variables

If X, Y bivariate Gaussian, $W_1 = a_1X + b_1Y$, $W_2 = a_2X + b_2Y$ are also bivariate Gaussian, with means/variances/covariance given by the usual bilinear formulas.

Example: $X \sim N[0, 16]$, $Z \sim N[0, 9]$ independent, $Y = X + Z$: $Y \sim N[0, 25]$ and $\rho_{X,Y} = \sqrt{(16/9)/(1 + 16/9)} = 4/5$.

Use it when / how: Use whenever a problem defines new variables as linear combinations of jointly Gaussian X, Y (e.g. $Y = X + Z$ signal-plus-noise) — you instantly know the result is Gaussian, so only the mean/variance/covariance remain to compute.

3 Chapter 6 — Functions of Random Variables

Theorem 6.1 — PMF of a Function of Two Discrete Random Variables (§6.1)

$$P_W(w) = \sum_{(x,y): g(x,y)=w} P_{X,Y}(x,y).$$

Example: Newsletter cost $W = LX$ (length $\times$ per-page rate): $W = 120$ arises from *two* different (l, x) pairs, $(2, 60)$ and $(3, 40)$, so $P_W(120) = P_{L,X}(2, 60) + P_{L,X}(3, 40)$ — add the probabilities of every input pair that maps to the same output value.

Use it when / how: The discrete analog of the Chapter 6 method: whenever several (x, y) pairs give the *same* $w = g(x, y)$, their probabilities *add*. Always check whether a given output value has more than one pre-image before writing down $P_W(w)$.

The General Method (§6.2)

To find $f_W(w)$ for $W = g(X)$: (1) find $F_W(w) = P[g(X) \leq w]$ by identifying which x satisfy $g(x) \leq w$; (2) differentiate, $f_W(w) = dF_W(w)/dw$.

Example: Pointer X uniform(0, 1) meters, $W = 100X$ cm: $F_W(w) = P[X \leq w/100] = F_X(w/100) = w/100$ on $(0, 100) \Rightarrow f_W(w) = 1/100$, i.e. $W \sim \text{Uniform}(0, 100)$.

Use it when / how: Use this whenever you need the full distribution of a transformed random variable — especially when g is not monotonic (e.g. $Y = X^2$, $Y = cX^2$, any limiter/clipper). If you only need $E[g(X)]$, use LOTUS (Theorem 4.4) instead — much faster.

Theorem 6.2 — Scaling: $W = aX$, $a > 0$

$$F_W(w) = F_X(w/a), \quad f_W(w) = \frac{1}{a} f_X(w/a).$$

Example: X triangular, $f_X(x) = 2x$ on $(0, 1)$; $W = aX \Rightarrow f_W(w) = \frac{1}{a} f_X(w/a) = 2w/a^2$ on $(0, a)$.

Use it when / how: Use for any pure rescaling of a random variable (unit conversions, gain factors).

Theorem 6.3 — Scaling Within Standard Families ($W = aX$, $a > 0$)

Uniform(b, c) $\rightarrow$ Uniform(ab, ac); Exponential(λ) $\rightarrow$ Exponential(λ/a); Erlang(n, λ) $\rightarrow$ Erlang($n, \lambda/a$);
Gaussian(μ, σ) $\rightarrow$ Gaussian($a\mu, a\sigma$).

Example: $X \sim \text{Exponential}(\lambda = 2)$, $W = 3X$: by the table, $W \sim \text{Exponential}(2/3)$ — confirmed by Theorem 6.2: $f_W(w) = \frac{1}{3}(2)e^{-2(w/3)} = \frac{2}{3}e^{-(2/3)w}$, rate $2/3$.

Use it when / how: Whenever $W = aX$ and X belongs to one of the four standard continuous families, skip Theorem 6.2's derivation entirely and just read the new parameter off this table.

Theorem 6.4 — Shift: $W = X + b$

$$F_W(w) = F_X(w - b), \quad f_W(w) = f_X(w - b).$$

Example: $X \sim \text{Uniform}(0, 1)$, $W = X - 2$: $f_W(w) = f_X(w + 2) = 1$ on $(-2, -1)$ — just the uniform density slid left by 2.

Use it when / how: Combine with Theorem 6.2 for the general affine case $W = aX + b$: $f_W(w) = \frac{1}{a} f_X(\frac{w-b}{a})$.

★ Squaring a Random Variable, e.g. $Y = cX^2$ (Example 6.4) ★ [PROFESSOR-FLAGGED]

Since Y can never be negative, “ Y is at most y ” means “ X is somewhere between $-\sqrt{y/c}$ and $+\sqrt{y/c}$.” So: rewrite the event that way, find its probability using X 's own density, then differentiate at the end to get Y 's density.

Example: Exam 2, Problem 3, worked in full: X is the current, Uniform($-2, 2$); $Y = 9X^2$ is the power. *Step 1:* since X is between -2 and 2 , $Y = 9X^2$ can only land between 0 and 36 . *Step 2:* “ $Y \leq y$ ” means “ $-\sqrt{y/9} \leq X \leq \sqrt{y/9}$.” Because X is uniform, that's just (length of that stretch)/(total length 4) = $\sqrt{y}/6$ — so $F_Y(y) = \sqrt{y}/6$. *Step 3:* take the derivative: $f_Y(y) = 1/(12\sqrt{y})$, for y between 0 and 36 .

Use it when / how: This is the exact recipe for every “power” or “energy” problem (“current X is uniform, power $Y = cX^2$, find the probability curve of Y ”) — it appeared almost word for word on Exam 2. In plain terms: (1) figure out the smallest and largest values Y could be, using the smallest/largest values X could be; (2) turn “ $Y \leq y$ ” into a statement about X being between two square-root bounds; (3) find that probability directly from X ’s density (watch out — if those square-root bounds ever stick out past where X is actually allowed to be, stop them at X ’s own edge instead); (4) differentiate to get Y ’s density.

Theorem 6.5 — Inverse-CDF Method

If $U \sim \text{Uniform}(0, 1)$ and F is an invertible CDF, then $X = F^{-1}(U)$ has CDF F .

Example: Target Exponential(1): solve $u = 1 - e^{-x} \Rightarrow x = -\ln(1 - u)$, so $X = -\ln(1 - U)$ is Exponential(1).

Use it when / how: Use to *construct* a random variable with a target distribution from a Uniform(0,1) generator (simulation-style questions). Fails for Gaussian (no closed-form inverse) and for mixed random variables.

★ A Meter That Can’t Read Past a Certain Value (Examples 6.7–6.8, §6.3) ★ [PROFESSOR-FLAGGED]

Below the cutoff, nothing changes — the density stays exactly what it was. At the cutoff itself, every probability that “would have gone higher” piles up into one single spot, sized to whatever was left over.

Example: Exam 2, Problem 3(c), worked in full: the power meter reads $W = Y$ normally, but shows 16 for anything 16 or above. *Step 1:* below 16, W behaves exactly like Y , so $F_W(w) = F_Y(w) = \sqrt{w}/6$ there — unchanged. *Step 2:* right at $w = 16$, the chance that used to be spread out above 16 (that’s $1 - F_Y(16) = 1 - 4/6 = 1/3$) all lands in one pile exactly at 16. *Result:* $f_W(w) = \frac{1}{12\sqrt{w}}$ for w below 16, plus one extra pile of size $\frac{1}{3}$ sitting exactly at $w = 16$.

Use it when / how: Use whenever a variable is described as “capped,” “limited,” or “can’t read past” some value (a power meter that maxes out, a voltage clipper, etc.) — this exact setup appeared on Exam 2, Problem 3(c). The result always has this same two-part shape: an ordinary curve below the cap, plus one single pile of leftover probability sitting right at the cap.

4 Chapter 13 — Stochastic Processes

4.1 Mean, Correlation, and the Stationarity Definitions (§13.7–13.8)

Definition 13.11/13.12/13.13 — Mean, Autocovariance, Autocorrelation

$$\mu_X(t) = E[X(t)]; \quad C_X(t, \tau) = \text{Cov}[X(t), X(t + \tau)]; \quad R_X(t, \tau) = E[X(t)X(t + \tau)].$$

Example: Random-phase carrier $X(t) = A \cos(2\pi f_c t + \Theta)$, $\Theta \sim \text{Uniform}(0, 2\pi)$ (Example 13.22): $\mu_X(t) = 0$ for all t , and $R_X(t, \tau) = (A^2/2) \cos(2\pi f_c \tau)$.

Use it when / how: These three functions are the *first* thing to compute for any named process. Note $R_X(t, 0) = E[X^2(t)]$ is the process's instantaneous average power.

Theorem 13.9 — Autocovariance from Autocorrelation

$$C_X(t, \tau) = R_X(t, \tau) - \mu_X(t)\mu_X(t + \tau).$$

Example: For the random-phase carrier (zero mean): $C_X(t, \tau) = R_X(t, \tau) - 0 = R_X(t, \tau)$ — mean and autocovariance coincide whenever $\mu_X = 0$.

Use it when / how: Use to convert between the two whenever a problem gives you one and asks for the other — identical in spirit to $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$ (Theorem 5.16(a)), just evaluated at two time instants.

Definition 13.14 — Stationary Process

Every joint PDF of $X(t_1), \dots, X(t_m)$ is unchanged under a common time shift τ .

Example: White Gaussian noise (Definition 13.20) is stationary: it's WSS by construction, and Theorem 13.15 (WSS + Gaussian $\Rightarrow$ stationary) upgrades that to full stationarity without checking every joint PDF by hand.

Use it when / how: The strict, hardest-to-verify definition — in practice you almost always check the easier WSS conditions (Definition 13.15) instead, since stationary $\Rightarrow$ WSS (Theorem 13.11).

★ Theorem 13.10 — Linear Transformations Preserve Stationarity ★ [PROFESSOR-FLAGGED]

$X(t)$ stationary $\Rightarrow Y(t) = aX(t) + b$ ($a > 0$) is also stationary.

Example: If $X(t)$ is a stationary noise voltage and an amplifier outputs $Y(t) = 5X(t) + 2$, then $Y(t)$ is automatically stationary too — no re-derivation needed.

Use it when / how: Use whenever a new process is defined as an affine function of a known stationary process — you get stationarity of $Y(t)$ for free, with no need to re-check every joint PDF.

★ Theorem 13.11 — Three Consequences of Stationarity ★ [PROFESSOR-FLAGGED]

Stationary $X(t) \Rightarrow$ (a) $\mu_X(t) = \mu_X$ (constant); (b) $R_X(t, \tau) = R_X(\tau)$ (τ only); (c) $C_X(t, \tau) = R_X(\tau) - \mu_X^2 = C_X(\tau)$.

Example: Random-phase carrier (Example 13.22): $\mu_X(t) = 0$ for all t (constant, satisfies (a)) and $R_X(t, \tau) = (A^2/2) \cos(2\pi f_c \tau)$ depends on τ only (satisfies (b)).

Use it when / how: This is your checklist for “is this process stationary/WSS.” Compute $\mu_X(t)$ and $R_X(t, \tau)$ directly from the process's definition, then check (a) and (b) in order — if either fails, the process is *not* stationary (and, since stationary $\Rightarrow$ WSS, likely not WSS either).

★ Example 13.22 — The Random-Phase AM Carrier ★ [PROFESSOR-FLAGGED]

$$X(t) = A \cos(2\pi f_c t + \Theta), \Theta \sim \text{Uniform}(0, 2\pi): \mu_X(t) = 0, \quad R_X(t, \tau) = \frac{A^2}{2} \cos(2\pi f_c \tau) = R_X(\tau).$$

Example: Worked in full: using $E[\cos(a + k\Theta)] = 0$ for any fixed a , nonzero integer k , expand $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ with $A - B = -2\pi f_c \tau$ constant — the $\cos(A + B)$ term (which contains 2Θ) averages to zero, leaving $R_X(t, \tau) = \frac{A^2}{2} \cos 2\pi f_c \tau$.

Use it when / how: The professor's flagship worked example — know the setup and result by heart. Use this exact template for *any* “random phase cosine/sinusoid” process question — constant zero mean and τ -only autocorrelation $\Rightarrow$ WSS (Example 13.23).

4.2 Wide Sense Stationarity & Ergodicity (§13.9)

Definition 13.15 — Wide Sense Stationary (WSS)

$E[X(t)] = \mu_X$ (constant) and $R_X(t, \tau) = R_X(\tau)$ (depends on τ only).

Example: The random-phase carrier of Example 13.22 has $\mu_X = 0$ (constant) and $R_X(\tau) = (A^2/2) \cos 2\pi f_c \tau$ (τ -only) $\Rightarrow$ WSS (Example 13.23).

Use it when / how: The practical, easy-to-check property you'll actually use on the exam (weaker than full stationarity, and every stationary process is automatically WSS by Theorem 13.11). Check these two conditions directly from $\mu_X(t)$ and $R_X(t, \tau)$ — don't bother with the full joint-PDF definition unless explicitly asked for strict-sense stationarity.

★ Theorem 13.12 — Properties of the WSS Autocorrelation Function ★ [PROFESSOR-FLAGGED]

$R_X(0) \geq 0$; $R_X(\tau) = R_X(-\tau)$ (even); $R_X(0) \geq |R_X(\tau)|$ (peak at $\tau=0$). *Bonus property used on the Ch. 13 homework:* if $X(t)$ has nonzero mean and no periodic component, $\lim_{|\tau| \rightarrow \infty} R_X(\tau) = \mu_X^2 \geq 0$.

Example: $R(\tau) = e^{-\tau^2} \sin \tau$ is **odd** (since $\sin$ is odd) $\Rightarrow$ violates $R(\tau) = R(-\tau) \Rightarrow$ **not** a valid autocorrelation function. Homework 13.9.1: $R_4(\tau) = \delta(\tau) - 10$ is even but $\lim_{|\tau| \rightarrow \infty} R_4(\tau) = -10 < 0$, impossible since this limit must equal $\mu_X^2 \geq 0 \Rightarrow$ **not valid**.

Use it when / how: Use this to (dis)prove a candidate function is a valid autocorrelation. Any function that is odd, or whose magnitude exceeds its value at $\tau = 0$, is automatically invalid; any function that approaches a *negative* constant as $|\tau| \rightarrow \infty$ is also invalid.

Definition 13.16 — Average Power

$$R_X(0) = E[X^2(t)].$$

Example: Random-phase carrier: $R_X(0) = (A^2/2) \cos(0) = A^2/2$ — the average power of the transmitted signal.

Use it when / how: Whenever a problem asks for “average power” of a WSS process/signal, this is just $R_X(0)$ — no extra computation needed once you have the autocorrelation function.

★ Ergodicity — Section 13.9's Named Concept ★ [PROFESSOR-FLAGGED]

A WSS process is **ergodic** if statistical (ensemble) averages equal the corresponding *time* averages of one sample function: $\langle X(t) \rangle = \lim_{T \rightarrow \infty} \bar{X}(T) = E[X(t)]$, and similarly $\langle X^2(t) \rangle = E[X^2(t)]$.

Example: $R_X(\tau) = 25 + \frac{4}{1+6\tau^2}$ (stationary, ergodic, no periodic part): $\mu_X^2 = \lim_{\tau \rightarrow \infty} R_X(\tau) = 25 \Rightarrow \mu_X = 5$; $E[X^2(t)] = R_X(0) = 29 \Rightarrow \text{Var}[X(t)] = 29 - 25 = 4$.

Use it when / how: This is exactly what the professor flagged as “Section 13.9.” Recognize an ergodicity question by phrases like “time average” vs. “ensemble/statistical average.” Formally, mean-ergodicity holds iff $\lim_{T \rightarrow \infty} \text{Var}[\bar{X}(T)] = 0$ (Theorem 13.13) — e.g. true whenever $\int_{-\infty}^{\infty} |C_X(\tau)| d\tau < \infty$, but **false** for $R(\tau) = 1 - |\tau|/T$ (never decays relative to the averaging window).

Theorem 13.13 — Mean-Ergodic Unbiased/Consistent Estimator

If $\int_{-\infty}^{\infty} |C_X(\tau)| d\tau < \infty$, then $\bar{X}(T)$ is unbiased ($E[\bar{X}(T)] = \mu_X$) and consistent ($\text{Var}[\bar{X}(T)] \rightarrow 0$ as $T \rightarrow \infty$).

Example: $R_X(\tau) = A\delta(\tau)$, zero mean: $\text{Var}[\bar{X}(T)] = A/T \rightarrow 0$ as $T \rightarrow \infty \Rightarrow$ mean-ergodic. Contrast: $R(\tau) = 1 - |\tau|/T$ gives $\text{Var}[\bar{X}_T] = 2/3 \neq 0 \Rightarrow$ **not** mean-ergodic.

Use it when / how: Use this integrability condition as the formal test for mean-ergodicity whenever you're given an explicit $C_X(\tau)$ or $R_X(\tau)$ and asked to justify (not just assert) ergodicity.

4.3 Cross-Correlation & Gaussian Processes (§13.10–13.11)**Definition 13.17/13.18 — Cross-Correlation & Jointly WSS**

$R_{XY}(t, \tau) = E[X(t)Y(t + \tau)]$. Jointly WSS: both individually WSS *and* $R_{XY}(t, \tau) = R_{XY}(\tau)$ depends only on τ .

Example: $Y(t) = X(t) + N(t)$, X, N independent WSS, $\mu_N = 0$: $R_{YX}(t, \tau) = R_X(\tau)$ (depends on τ only) $\Rightarrow X(t), Y(t)$ jointly WSS.

Use it when / how: Use whenever two processes are combined (e.g. signal plus independent noise, $Y(t) = X(t) + N(t)$) and you're asked whether they are individually WSS, and whether they are *jointly* WSS — these are two separate checks that must *both* be verified from the definitions, since one can hold without the other.

Theorem 13.14 — Symmetry of Cross-Correlation

If $X(t), Y(t)$ jointly WSS: $R_{XY}(\tau) = R_{YX}(-\tau)$.

Example: For $Y(t) = X(t) + N(t)$ (Example 13.24): $R_{YX}(\tau) = R_X(\tau)$, so Theorem 13.14 gives $R_{XY}(\tau) = R_{YX}(-\tau) = R_X(-\tau) = R_X(\tau)$ by evenness (Theorem 13.12) — consistent with X, Y being jointly WSS.

Use it when / how: Use to convert between R_{XY} and R_{YX} — they are reflections of each other about $\tau = 0$, not generally equal to each other directly.

★ Technique — Product of Two Independent WSS Processes (Homework 13.10.1) ★ [PROFESSOR-FLAGGED]

If $X(t), Y(t)$ are independent WSS with means μ_X, μ_Y and autocorrelations $R_X(\tau), R_Y(\tau)$, and $W(t) = X(t)Y(t)$, then $\mu_W = \mu_X\mu_Y$ (constant) and $R_W(t, \tau) = R_X(\tau)R_Y(\tau)$ (τ only) $\Rightarrow W(t)$ is WSS. Also $R_{WX}(t, \tau) = R_X(\tau)\mu_Y$ (τ only) $\Rightarrow W(t), X(t)$ jointly WSS.

Example: Homework 13.10.1: split $E[W(t)W(t + \tau)] = E[X(t)X(t + \tau)Y(t)Y(t + \tau)]$ into $E[X(t)X(t + \tau)] \cdot E[Y(t)Y(t + \tau)] = R_X(\tau)R_Y(\tau)$ using independence of the X and Y processes — the same “split the expectation of a product using independence” trick as Theorem 5.17(a), just applied to two processes evaluated at two time instants each.

Use it when / how: Use whenever a new process is defined as the *product* of two independent WSS processes (as opposed to their sum, Definition 13.17/13.18's example) — multiply the individual autocorrelations to get the new process's autocorrelation, and check individual-vs-joint WSS exactly as with a sum.

Definition 13.19 — Gaussian Process

$X(t)$ is Gaussian iff every finite sample vector $[X(t_1) \cdots X(t_k)]'$ is a Gaussian random vector.

Example: White Gaussian noise $W(t)$: any collection $W(t_1), \dots, W(t_k)$ is a Gaussian random vector (in fact independent Gaussians, since uncorrelated + jointly Gaussian $\Rightarrow$ independent).

Use it when / how: Recognize this whenever a process is built from Gaussian noise/signals via linear operations — linear combinations of jointly Gaussian variables stay Gaussian (Theorem 5.21), so such processes are automatically Gaussian processes.

Theorem 13.15 — A WSS Gaussian Process Is Stationary

WSS + Gaussian $\Rightarrow$ (strict-sense) stationary.

Example: If a Gaussian noise process is shown to have constant mean and τ -only autocorrelation (i.e. WSS), this theorem immediately upgrades that to full stationarity — no further joint-PDF checking required.

Use it when / how: A major shortcut: for Gaussian processes specifically, you never need to separately verify full stationarity — just check the two easy WSS conditions (constant mean, τ -only autocorrelation) and stationarity follows automatically.

★ Bivariate Special Case for a Gaussian Process ★ [PROFESSOR-FLAGGED]

For a stationary Gaussian process, the joint PDF of $X_1 = X(t)$, $X_2 = X(t+\tau)$ is exactly Definition 5.10's bivariate Gaussian PDF with $\sigma_1 = \sigma_2 = \sqrt{R_X(0)}$ and $\rho = C_X(\tau)/R_X(0)$.

Example: $\mu_X = 0$, $R_X(\tau) = 2^{-|\tau|}$: $\text{Var}[X(t)] = R_X(0) = 1$, $\rho = R_X(1)/R_X(0) = 1/2 \Rightarrow f_{X(t), X(t+1)}(x_1, x_2) = \frac{1}{\pi\sqrt{3}} \exp\left(-\frac{2(x_1^2 - x_1x_2 + x_2^2)}{3}\right)$.

Use it when / how: Use whenever you're given $R_X(\tau)$ for a stationary Gaussian process and asked for the joint PDF of two samples: compute $\sigma^2 = R_X(0)$ and $\rho = C_X(\tau)/R_X(0)$, then plug directly into the bivariate Gaussian formula (Definition 5.10) — no need to re-derive anything from the general n -dimensional Gaussian vector PDF.

Definition 13.20 — White Gaussian Noise

Stationary Gaussian process, $\mu_W = 0$, $R_W(\tau) = \eta_0\delta(\tau)$.

Example: Average power $R_W(0) = \eta_0\delta(0) = \infty$ — physically impossible but useful as an idealization for filtered noise with finite power.

Use it when / how: Recognize this whenever “white noise” is mentioned: samples at any two distinct times are automatically *independent* (uncorrelated + jointly Gaussian $\Rightarrow$ independent, Theorem 5.20), and average power $R_W(0) = \infty$ (never physically realizable).

Part I

Exam 1, Exam 2 & Chapter 13 Homework —
Every Problem, Solved

5 Exam 1 — All 7 Problems

Exam 1, Problem 1 — Mixed random variable

Theorems/tools used: Definition 4.12 (delta function), Theorem 4.16 (sifting), Theorem 4.2 (normalization), Theorem 4.1 (CDF from jumps)

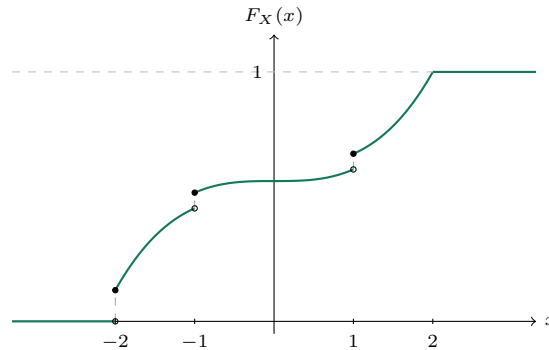
$$f_X(x) = \frac{1}{8}\delta(x+2) + \frac{1}{16}\delta(x+1) + \frac{1}{16}\delta(x-1) + Ax^2 \text{ on } |x| \leq 2.$$

Step 1 (Theorem 4.2 — normalization forces $\int f_X = 1$): the three impulses already contribute a fixed amount of probability, $\frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4}$, so whatever is left, $1 - \frac{1}{4} = \frac{3}{4}$, must come from the continuous piece. Setting $\int_{-2}^2 Ax^2 dx = A \cdot \frac{16}{3}$ equal to $\frac{3}{4}$ and solving gives $A = 9/64$.

Step 2 (Theorem 4.1(c) + Theorem 4.17 — build F_X from jumps): integrate the continuous density piecewise and add each impulse's full weight the instant x crosses it (that's exactly what Theorem 4.17's $\delta = du/dx$ identity encodes):

$$F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{2} + \frac{3x^3}{64} & -2 \leq x < -1 \\ \frac{9}{16} + \frac{3x^3}{64} & -1 \leq x < 1 \\ \frac{5}{8} + \frac{3x^3}{64} & 1 \leq x < 2 \\ 1 & x \geq 2. \end{cases}$$

Part (b) — the requested sketch of $F_X(x)$: plot the smooth cubic pieces and mark each impulse as a vertical jump (open circle = limit from the left, filled circle = the right-continuous value actually attained):



Step 3 (Theorem 4.1(c) — interval probability from the CDF): read off $F_X(1) = 43/64$ directly, and $P[-1 < X \leq 2] = F_X(2) - F_X(-1) = 1 - 33/64 = 31/64$.

Exam 1, Problem 2 — Joint PDF, marginals, moments

Theorems/tools used: Definition 5.5-style marginalization, Theorem 5.9 ($E[g(X, Y)]$), Theorem 4.5(c) (variance from moments)

$$f_{X,Y}(x, y) = \frac{x+y}{3} \text{ on } 0 \leq x \leq 1, 0 \leq y \leq 2.$$

Step 1 (marginalize — integrate out y): $f_X(x) = \int_0^2 \frac{x+y}{3} dy = \frac{2x+2}{3}$ on $0 \leq x \leq 1$.

Step 2 (Theorem 5.9/LOTUS on the marginal): treat f_X as an ordinary single-variable PDF and apply Theorem 4.4: $E[X] = \int_0^1 x \cdot \frac{2x+2}{3} dx = 5/9$, $E[X^2] = \int_0^1 x^2 \cdot \frac{2x+2}{3} dx = 7/18$.

Step 3 (Theorem 4.5(c) — variance): $\text{Var}[X] = 7/18 - (5/9)^2 = 13/162$.

Exam 1, Problem 3 — Joint PMF table

Theorems/tools used: Marginal PMF (sum the joint table over rows/columns), Definition 4.4/Theorem 4.5(c) (mean/variance from a PMF)

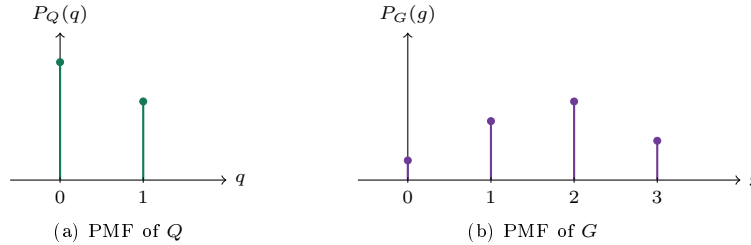
Joint PMF $P_{Q,G}(q,g)$ with row sums $P_Q(0) = 0.6$, $P_Q(1) = 0.4$ and column sums $P_G(0)=0.1$, $P_G(1)=0.3$, $P_G(2)=0.4$, $P_G(3)=0.2$.

Step 1 (event probabilities — sum the joint PMF over the relevant cells): $P[Q=0] = 0.6$ (a marginal, read directly); $P[Q=G] = P_{Q,G}(0,0) + P_{Q,G}(1,1) = 0.18$; $P[G=0] = 0.1$; $P[G>1] = 0.6$; $P[G>Q] = 0.78$.

Step 2 (Definition 4.4 on the marginal P_Q): $E[Q] = 0(0.6) + 1(0.4) = 0.4$.

Step 3 (same idea on the marginal P_G , then Theorem 4.5(c)): $E[G] = 1.7$, $E[G^2] = 3.7 \Rightarrow \boxed{\text{Var}[G] = 3.7 - 1.7^2 = 0.81}$.

Requested sketches of the two marginal PMFs (each a set of spikes/impulses at integers, height = probability):

**Exam 1, Problem 4 — Normalize a continuous PDF**

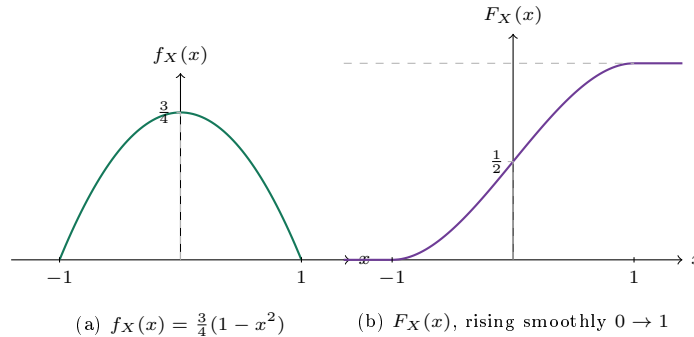
Theorems/tools used: Theorem 4.2 (normalization), Theorem 4.1/4.3 (CDF and interval probability)

$f_X(x) = c(1-x^2)$ on $[-1, 1]$.

Step 1 (Theorem 4.2 — set $\int f_X = 1$): $c \int_{-1}^1 (1-x^2) dx = \frac{4}{3}c = 1 \Rightarrow \boxed{c = 3/4}$.

Step 2 (Theorem 4.2(b) — integrate to get the CDF): $F_X(x) = \frac{3}{4} \left[(x+1) - \frac{1}{3}(x^3+1) \right]$.

Requested sketches of f_X and F_X :



Step 3 (apply Theorem 4.3/4.1(c)): $P[X=0] = 0$ (continuous r.v., any single point has zero probability); $F_X(0) = 1/2$; $P[0 < X < 0.5] = F_X(0.5) - F_X(0) = 11/32$; $P[|X - \frac{1}{2}| < \frac{1}{4}] = F_X(3/4) - F_X(1/4) \approx 0.2734$.

Exam 1, Problem 5 — Gaussian noise, Q -function, SNR

Theorems/tools used: Definition 4.11 (Q -function), Theorem 4.14 (Gaussian CDF via standardizing)

Bit-error probability $= P[X > \sqrt{E}] = Q(\sqrt{E/\sigma^2})$, $\text{SNR} = E/\sigma^2$.

Step 1 (Theorem 4.14 — standardize, then Definition 4.11): with $\text{SNR} = 4$, $Q(\sqrt{4}) = Q(2) = 1 - \Phi(2) = 1 - 0.97725 = \boxed{0.02275}$.

Step 2 (invert the same relationship): need $Q(\sqrt{\text{SNR}}) < 10^{-6}$; from the Q -table this requires $\sqrt{\text{SNR}} \geq 4.75 \Rightarrow \boxed{\text{SNR} \geq 22.56}$.

Exam 1, Problem 6 (Extra Credit) — Joint CDF normalizing constant

Theorems/tools used: The defining CDF limit $F_{X,Y}(\infty, \infty) = 1$ (the 2-D analog of Theorem 4.1(b))

$$F_{X,Y}(x,y) = a \left[\frac{\pi}{2} + \tan^{-1}(x/2) \right] \left[\frac{\pi}{2} + \tan^{-1}(y/3) \right].$$

Step (take the limit as $x, y \rightarrow \infty$ and set it to 1): since $\tan^{-1}(\infty) = \pi/2$, each bracket $\rightarrow \pi$, so $\lim F_{X,Y} = a\pi \cdot \pi = 1 \Rightarrow a = 1/\pi^2$. (Checked: as $x, y \rightarrow -\infty$ each bracket $\rightarrow 0$, so $F(-\infty, y) = F(x, -\infty) = 0$ as required of any joint CDF.)

Exam 1, Problem 7 (Extra Credit) — Soft limiter, Jacobian method

Theorems/tools used: The Chapter 6 general method (§6.2), applied via the single-variable transformation formula $f_Y(y) = f_X(x)|dx/dy|$ on each monotonic branch separately

$Y = V(1 - e^{-X/a})$ for $X \geq 0$; $Y = -V(1 - e^{X/a})$ for $X \leq 0$ — two separate monotonic branches, so invert each one.

Step 1 (invert the $X \geq 0$ branch): $y = V(1 - e^{-x/a}) \Rightarrow x = a \ln\left(\frac{V}{V-y}\right)$, giving $dx/dy = a/(V-y)$.

Step 2 (invert the $X \leq 0$ branch the same way): $x = -a \ln\left(\frac{V}{V+y}\right)$, giving $dx/dy = a/(V+y)$.

Step 3 (combine both branches with unit-step indicators $u(y), u(-y)$ to cover the whole real line):

$$f_Y(y) = \frac{a}{V-y} f_X\left[a \ln\left(\frac{V}{V-y}\right)\right] u(y) + \frac{a}{V+y} f_X\left[-a \ln\left(\frac{V}{V+y}\right)\right] u(-y).$$

6 Exam 2 — All 5 Problems

Exam 2, Problem 1 — Joint PDF moments and covariance (textbook 5.7.12)

Theorems/tools used: Theorem 5.9 ($E[XY]$), Theorem 5.16(a) (covariance), Theorem 5.12 (variance of a sum)

$f_{X,Y}(x,y) = 4xy$ on the unit square.

Step 1 (Theorem 5.9 — integrate x , x^2 , y , y^2 , and xy against the joint PDF): $E[X] = E[Y] = 2/3$; $E[X^2] = E[Y^2] = 1/2 \Rightarrow$ (Theorem 4.5(c)) $\text{Var}[X] = \text{Var}[Y] = 1/18$; and $E[XY] = 4/9$.

Step 2 (Theorem 5.16(a) — $\text{Cov}[X,Y] = E[XY] - E[X]E[Y]$): $\text{Cov}[X,Y] = 4/9 - (2/3)^2 = 0$.

Step 3 (Theorem 5.11 then Theorem 5.12): $E[X+Y] = 4/3$, and since the covariance term vanishes, $\text{Var}[X+Y] = 1/18 + 1/18 + 2(0) = 1/9$.

Exam 2, Problem 2 — Bivariate-Gaussian-type joint PDF (textbook 5.9.5)

Theorems/tools used: Definition 5.10 (bivariate Gaussian PDF — coefficient matching), Theorem 5.19 (ρ is the correlation coefficient), Theorem 5.20 (uncorrelated $\Leftrightarrow$ independent)

$f_{X,Y}(x,y) = ce^{-(2x^2 - 4xy + 4y^2)}$.

Step 1 (recognize the form and match to Definition 5.10): write the target exponent $-\frac{1}{2(1-\rho^2)} \left[\frac{x^2}{\sigma_X^2} - \frac{2\rho xy}{\sigma_X \sigma_Y} + \frac{y^2}{\sigma_Y^2} \right]$ against $-(2x^2 - 4xy + 4y^2)$ and equate coefficients of x^2, xy, y^2 . Since there's no linear term, $E[X] = E[Y] = 0$.

Step 2 (solve the coefficient-matching system): this yields $\sigma_X = 1/\sqrt{2}$, $\sigma_Y = 1/2$, $\rho = 1/\sqrt{2}$.

Step 3 (Definition 5.10's normalizing constant): $c = 1/(2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}) = 2/\pi$.

Step 4 (Theorem 5.20): since $\rho \neq 0$ and X, Y are bivariate Gaussian, uncorrelated would have been equivalent to independent — but $\rho \neq 0$ means **dependent**.

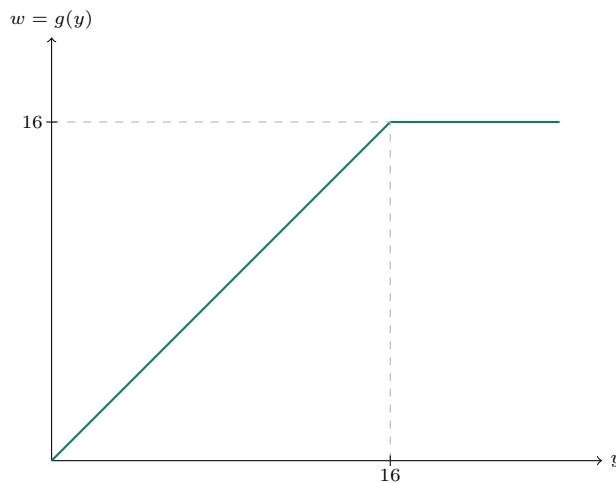
Exam 2, Problem 3 — Power dissipation & range-limited output (textbook 6.3.10)

Theorems/tools used: The $Y = cX^2$ template (Example 6.4/§6.2), the range-limited/clipped-variable impulse technique (Examples 6.7–6.8/§6.3)

$X \sim \text{Uniform}(-2, 2)$ (current), $Y = 9X^2$ (power, so $c = 9$).

Step 1 (the $F_W \rightarrow f_W$ method, §6.2, with the $Y = cX^2$ clipping refinement): $0 \leq Y < 36$; for $0 \leq y < 36$, $F_Y(y) = P[-\sqrt{y/9} \leq X \leq \sqrt{y/9}] = \sqrt{y}/6$ (both bounds fall inside X 's support here, so no clipping is needed) $\Rightarrow$ differentiate: $f_Y(y) = 1/(12\sqrt{y})$.

Step 2 (Examples 6.7–6.8's range-limiting technique — part (c)): $W = Y$ if $Y < 16$, else 16. It helps to first sketch W as a deterministic function of Y (the problem's own suggestion):



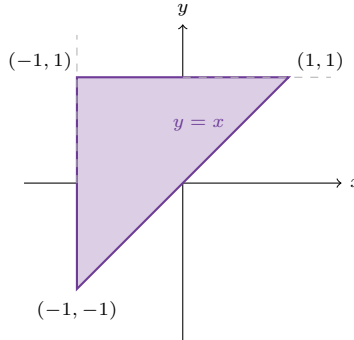
For $0 \leq w < 16$, $F_W(w) = F_Y(w) = \sqrt{w}/6$ (unchanged, limiter inactive). At $w = 16$ the CDF jumps from $F_Y(16^-) = 4/6$ to 1 — an impulse of weight $1 - 4/6 = 1/3$ appears in the PDF exactly at the clip point: $f_W(w) = \frac{1}{12\sqrt{w}} (0 \leq w < 16) + \frac{1}{3}\delta(w-16)$.

Exam 2, Problem 4 — Joint PDF on a triangle (textbook 5.8.8)

Theorems/tools used: Theorem 5.9 ($E[g(X, Y)]$ for $g = xy$ and $g = e^{x+y}$), Definition 5.7 (correlation)

$f_{X,Y}(x, y) = \frac{1}{2}$ on the triangle $-1 \leq x \leq y \leq 1$.

Requested sketch of the support region (shaded), bounded by $x = -1$, $y = 1$, and the line $y = x$:



Step 1 (Definition 5.7 via Theorem 5.9 — integrate xy over the triangle): $r_{X,Y} = E[XY] = \int_{-1}^1 \int_x^1 \frac{xy}{2} dy dx = \frac{1}{4} \int_{-1}^1 x(1 - x^2) dx = \boxed{0}$ (the integrand is odd in x over a symmetric range).

Step 2 (Theorem 5.9 again, this time with $g(x, y) = e^{x+y}$): $E[e^{X+Y}] = \int_{-1}^1 \int_x^1 \frac{1}{2} e^x e^y dy dx = \frac{1}{2} \int_{-1}^1 e^x (e - e^x) dx = \boxed{e^2/4 + e^{-2}/4 - 1/2}$.

Exam 2, Problem 5 — Discrete joint PMF, full moment workup (textbook 5.8.6)

Theorems/tools used: Marginal PMF (sum the joint table), Definition 5.7 (correlation), Theorem 5.16(a) (covariance), Definition 5.6 (correlation coefficient)

$P_{X,Y}(x, y) = 1/(4x)$ for integers $1 \leq y \leq x \leq 4$.

Step 1 (marginalize, then Definition 4.4/Theorem 4.5(c) on each marginal): $E[X] = 5/2, E[Y] = 7/4; \text{Var}[X] = 5/4, \text{Var}[Y] = 41/48$.

Step 2 (Definition 5.7 via Theorem 5.9 — sum $xy P_{X,Y}(x, y)$ over the table): $r_{X,Y} = E[XY] = 5$.

Step 3 (Theorem 5.16(a)): $\text{Cov}[X, Y] = 5 - (5/2)(7/4) = 10/16$.

Step 4 (Definition 5.6 — divide by $\sigma_X \sigma_Y$): $\rho_{X,Y} = \frac{10/16}{\sqrt{(5/4)(41/48)}} \approx 0.605$.

7 Chapter 13 Homework — All 6 Problems

Problem 13.9.1 — Which are valid autocorrelation functions?

Theorems/tools used: Theorem 13.12 (even, $R(0) \geq |R(\tau)|$), the extra limit property $\lim_{|\tau| \rightarrow \infty} R_X(\tau) = \mu_X^2 \geq 0$

$R_1(\tau) = \delta(\tau)$, $R_2(\tau) = \delta(\tau) + 10$, $R_3(\tau) = \delta(\tau - 10)$, $R_4(\tau) = \delta(\tau) - 10$.

Step 1 (Theorem 13.12(b) — check evenness first, it's the fastest filter): R_1, R_2, R_4 are all even; $R_3(-\tau) = \delta(\tau + 10) \neq R_3(\tau)$ — R_3 is NOT valid.

Step 2 (R_1 — recognize the white-noise form, Definition 13.20): $\delta(\tau)$ is exactly $R_W(\tau) = \eta_0 \delta(\tau)$ with $\eta_0 = 1$. Valid.

Step 3 (R_2 — decompose as a sum of two valid ACFs): $\delta(\tau) + 10 = (\text{white noise ACF}) + (\text{ACF of an independent random-constant process with } E[A^2] = 10)$; sums of independent processes' ACFs are valid ACFs. Valid.

Step 4 (R_4 — apply the extra limit property): even, but $\lim_{|\tau| \rightarrow \infty} R_4(\tau) = -10$, which must equal $\mu_X^2 \geq 0$ — contradiction. NOT valid.

Problem 13.9.5 — Autocorrelation of an iid sequence

Theorems/tools used: Definition 13.13 ($R_X[n, k] = E[X_n X_{n+k}]$), Theorem 4.5(c), Theorem 5.17(b) (independence)

X_n iid, $E[X_n] = \mu$, $\text{Var}[X_n] = \sigma^2$.

Step 1 ($k = 0$, Theorem 4.5(c)): $R_X[n, 0] = E[X_n^2] = \text{Var}[X_n] + \mu^2 = \sigma^2 + \mu^2$.

Step 2 ($k \neq 0$, independence + Theorem 5.17(b)): X_n, X_{n+k} are independent $\Rightarrow R_X[n, k] = E[X_n] E[X_{n+k}] = \mu^2$.

$$R_X[n, k] = \begin{cases} \sigma^2 + \mu^2 & k = 0 \\ \mu^2 & k \neq 0 \end{cases}$$

Problem 13.9.9 — WSS process times a random-phase cosine

Theorems/tools used: Definition 13.16 (average power), Example 13.22's lemma, Theorem 5.17(a) (independence)

$X(t)$ WSS, $R_X(0) = E[X^2(t)] = 1$; $\Theta \sim \text{Uniform}(0, 2\pi)$ independent of $X(t)$.

(a) Definition 13.16 directly: $E[X^2(t)] = R_X(0) = \boxed{1}$.

(b) Example 13.22's lemma with $a = 2\pi f_c t$, $k = 1$: $E[\cos(2\pi f_c t + \Theta)] = \boxed{0}$.

(c) Theorem 5.17(a) on $Y(t) = X(t) \cos(2\pi f_c t + \Theta)$: $E[Y(t)] = E[X(t)] \cdot E[\cos(2\pi f_c t + \Theta)] = E[X(t)] \cdot 0 = \boxed{0}$.

(d) Theorem 5.17(a) again, then the lemma with $k = 2$: $E[Y^2(t)] = E[X^2(t)] \cdot E[\cos^2(2\pi f_c t + \Theta)] = 1 \cdot \left[\frac{1}{2} + \frac{1}{2} E[\cos(4\pi f_c t + 2\Theta)]\right] = 1 \cdot \frac{1}{2} = \boxed{1/2}$.

Problem 13.10.1 — Product of two independent WSS processes

Theorems/tools used: The product-of-independent-WSS-processes technique (this page's flagged entry above), Theorem 5.17(a)

$X(t), Y(t)$ independent WSS, means μ_X, μ_Y , autocorrelations $R_X(\tau), R_Y(\tau)$; $W(t) = X(t)Y(t)$.

(a) Mean and autocorrelation, via independence: $\mu_W = \mu_X \mu_Y$; $R_W(t, \tau) = E[X(t)X(t+\tau)] \cdot E[Y(t)Y(t+\tau)] = R_X(\tau)R_Y(\tau)$ — τ -only $\Rightarrow$ $W(t)$ is WSS.

(b) Cross-correlation with $X(t)$: $R_{WX}(t, \tau) = E[X(t)X(t+\tau)] \cdot E[Y(t)] = R_X(\tau)\mu_Y$ — τ -only $\Rightarrow$ yes, jointly WSS.

Problem 13.10.3 — Autocorrelation/cross-correlation of a delayed process

Theorems/tools used: Definition 13.13 (autocorrelation), Definition 13.17/13.18 (cross-correlation, jointly WSS)

$X(t)$ WSS, $R_X(\tau) = 10 \sin(2000\pi\tau)/(2000\pi\tau)$; $Y(t) = X(t - t_0)$, $t_0 = 5 \times 10^{-5}$ s.

(a) The two arguments of X differ by τ regardless of t_0 : $R_Y(\tau) = R_X(\tau)$ — delaying a WSS process doesn't change its ACF.

- (b) The two arguments now differ by $\tau - t_0$: $R_{XY}(\tau) = R_X(\tau - t_0)$.
- (c) $R_Y(\tau)$ depends on τ only and $\mu_Y = \mu_X$ (constant) $\Rightarrow Y(t)$ is WSS.
- (d) $R_{XY}(t, \tau) = R_X(\tau - t_0)$ depends on τ only $\Rightarrow$ jointly WSS.

Problem 13.11.2 — Which transforms of a Gaussian process stay Gaussian?

Theorems/tools used: Definition 13.19 (Gaussian process), Theorem 5.21/8.11 (linear transforms of Gaussian vectors are Gaussian)

Given Gaussian $X(t)$: (a) $2X(t)$, (b) $X(t/2)$, (c) $X(t)/2$, (d) $X(t) - X(t-1)$, (e) $X(2t)$.

Step 1 (scalar multiples, (a) and (c)): $[2X(t_1), \dots, 2X(t_k)]'$ and $[X(t_1)/2, \dots, X(t_k)/2]'$ are linear (scalar) transforms of a Gaussian vector. Gaussian.

Step 2 (time re-indexing, (b) and (e)): $[X(t_1/2), \dots, X(t_k/2)]'$ and $[X(2t_1), \dots, X(2t_k)]'$ are just samples of X at different (still arbitrary) time instants — Gaussian by Definition 13.19 directly. Gaussian.

Step 3 (linear combination across two times, (d)): $[X(t_1) - X(t_1 - 1), \dots, X(t_k) - X(t_k - 1)]'$ is a linear transform of the $2k$ -dimensional Gaussian vector $[X(t_1), X(t_1 - 1), \dots, X(t_k), X(t_k - 1)]'$. Gaussian.

All five (a)–(e) are Gaussian processes.

One-Page Recap: How Each Exam Problem Is Solved — Theorem & Technique at a Glance

This table is the theoretical skeleton behind every solution in Part I: for each problem, *which theorem* supplies the formula and *which technique* turns that formula into the answer. Study this table first — it is the pattern you are expected to reproduce on the final for a similarly-worded question.

Problem	Theorem(s) used	Technique — <i>how</i> it's solved
Exam 1, P1 (mixed r.v.)	Def. 4.12, Thm. 4.16, Thm. 4.2, Thm. 4.1(c)/4.17	Normalize-then-integrate: impulse probabilities are fixed numbers, so subtract their total from 1 to get the continuous part's required area; solve for the constant. Then build F_X by integrating the continuous density piecewise and adding each impulse's full weight as a jump the instant x crosses it.
Exam 1, P2 (joint PDF $\rightarrow$ marginal $\rightarrow$ moments)	Marginalization, Thm. 4.4 (LO-TUS), Thm. 4.5(c)	Reduce to one variable: integrate the joint PDF over the <i>other</i> variable to get $f_X(x)$ alone, then solve it as an ordinary single-variable problem ($E[X]$, $E[X^2]$, subtract the square for variance).
Exam 1, P3 (joint PMF table)	Marginal PMF (row/column sums), Def. 4.4, Thm. 4.5(c)	Sum the table: row sums give one marginal, column sums the other; event probabilities are sums of the specific cells satisfying the event; moments follow from the marginal exactly as in the continuous case, just with sums instead of integrals.
Exam 1, P4 (normalize a PDF)	Thm. 4.2, Thm. 4.1/4.3	Set $\int f_X = 1$ and solve for the constant; integrate once more to get F_X ; every subsequent probability is then a plug-in evaluation or difference of F_X .
Exam 1, P5 (Gaussian/ Q -function/SNR)	Def. 4.11, Thm. 4.14	Standardize, then table-lookup: rewrite the event in terms of $Q(z)$ or $\Phi(z)$ with z a function of the given parameter (here $\sqrt{\text{SNR}}$); to go the other direction (part (b)), invert the Q -table to find the z that achieves the target probability.
Exam 1, P6 (joint CDF constant)	$F_{X,Y}(\infty, \infty) = 1$	Take the defining limit: let $x, y \rightarrow \infty$ in the given formula (using known limits like $\tan^{-1}(\infty)$), set the result equal to 1, and solve for the constant.
Exam 1, P7 (soft limiter)	Ch. 6 general method, $f_Y = f_X dx/dy $	Invert each monotonic branch separately: split the domain where g is monotonic, solve $y = g(x)$ for x on each piece, multiply f_X at that x by $ dx/dy $, and stitch the pieces together with unit-step indicators.
Exam 2, P1 (joint PDF moments/covariance)	Thm. 5.9, Thm. 5.16(a), Thm. 5.12	Moments first, combine second: compute $E[X]$, $E[Y]$, $E[X^2]$, $E[Y^2]$, $E[XY]$ all by direct double integration (Thm. 5.9), then get $\text{Cov}[X, Y]$ and $\text{Var}[X + Y]$ purely algebraically from those numbers — never integrate $(x - \mu_X)(y - \mu_Y)$ directly.
Exam 2, P2 (bivariate-Gaussian-type PDF)	Def. 5.10, Thm. 5.19, Thm. 5.20	Coefficient matching: recognize the $ce^{-(\text{quadratic})}$ form, equate its quadratic's coefficients term-by-term with Definition 5.10's exponent to solve for σ_X, σ_Y, ρ simultaneously, then get c from the normalizing constant; use $\rho \neq 0$ to conclude dependence via Theorem 5.20.

Problem	Theorem(s) used	Technique — <i>how</i> it's solved
Exam 2, P3 (power dissipation, range-limited)	$Y = cX^2$ template (Ex. 6.4), clip-impulse technique (Ex. 6.7–6.8)	Two-stage transform: first derive F_Y by translating $\{Y \leq y\}$ into a symmetric interval on X and integrating (clipping to X 's support if needed); then, for the range-limited version, keep f_Y unchanged below the clip and add an impulse at the clip point weighted by the clipped-away tail probability.
Exam 2, P4 (joint PDF on a triangle)	Thm. 5.9, Def. 5.7	Set up the double integral over the triangular support (inner limit depends on the outer variable), then evaluate directly for whichever function $g(x, y)$ (xy , or e^{x+y}) is asked for.
Exam 2, P5 (discrete joint PMF workup)	Marginal PMF, Def. 5.7, Thm. 5.16(a), Def. 5.6	Same pipeline as P1, discrete version: marginalize by summing the table, get $E[XY]$ by summing $xy P_{X,Y}(x, y)$ over all cells, then chain into covariance and $\rho_{X,Y}$ algebraically.
HW 13.9.1 (valid ACF?)	Thm. 13.12, limit property $\lim_{ \tau \rightarrow \infty} R_X(\tau) = \mu_X^2$	Test each candidate against Theorem 13.12: check evenness first (fastest rejection), then $R(0) \geq R(\tau) $, then the limit-as- $ \tau \rightarrow \infty$ property; a candidate that's a sum of two individually-valid ACFs is itself valid.
HW 13.9.5 ($R_X[n, k]$ of an iid sequence)	Def. 13.13, Thm. 4.5(c), Thm. 5.17(b)	Split on $k = 0$ vs. $k \neq 0$: at $k = 0$ use $E[X_n^2] = \text{Var}[X_n] + \mu^2$; at $k \neq 0$ use independence to factor $E[X_n X_{n+k}] = E[X_n] E[X_{n+k}]$.
HW 13.9.9 (WSS process $\times$ random-phase cosine)	Def. 13.16, Example 13.22's lemma, Thm. 5.17(a)	Factor every expectation using independence of $X(t)$ and Θ , then apply Example 13.22's $E[\cos(a + k\Theta)] = 0$ lemma to kill the oscillating terms.
HW 13.10.1 (product of independent WSS processes)	The product-of-independent-processes technique, Thm. 5.17(a)	Split $E[W(t)W(t+\tau)]$ into $E[X(t)X(t+\tau)] \cdot E[Y(t)Y(t+\tau)] = R_X(\tau)R_Y(\tau)$ using independence of the two processes — multiply autocorrelations instead of adding them (contrast with the signal-plus-noise <i>sum</i> template).
HW 13.10.3 (delayed process, cross-correlation)	Def. 13.13, Def. 13.17/13.18	Track the time difference between the two arguments of X: for R_Y it stays τ (delay cancels); for R_{XY} it becomes $\tau - t_0$ — read the answer straight off R_X evaluated at that difference.
HW 13.11.2 (Gaussian process under transforms)	Def. 13.19, Thm. 5.21/8.11	Two closure facts do all five parts: scalar multiples and linear combinations of Gaussian samples stay Gaussian (Thm. 5.21/8.11), and re-indexing time (even nonlinearly, e.g. $t \mapsto 2t$) still just samples X at particular instants (Def. 13.19).

Exam Pattern Analysis: What the Professor Actually Tests, in Plain Language

Every problem from Exam 1, Exam 2, and the Chapter 13 homework was sorted below into one of nine “types.” For each type you get: how often it showed up, what you’re really being asked to do (in plain words), and the steps — with the actual numbers from a real problem attached to each step, so you can see exactly how the idea turns into an answer.

The Headline Finding

Of the **12** Chapter 4–6 exam problems, **5 (42%)** are all the same underlying task: boil a two-variable problem down to a few plain numbers, then combine those numbers with simple arithmetic (Type 2 below) — by far the most-repeated skill on both exams. Of the **6** Chapter 13 homework problems, **3 (50%)** are all about splitting an average because two things don't affect each other (Type 8 below). **If you only have time to practice two things before the final, practice these two.**

Chapters 4–6: Six Problem Types (from Exam 1 & Exam 2)

Type 1 — Solve for the Missing Number, Then Build the Full Probability Curve

Seen in: Exam 1, Problems 1 and 4 (2 of 12)

Example: You're given a probability curve — the PDF, $f_X(x)$ — with one unknown number sitting in it. Pin down that number first, then use it to write out the complete “chance X is at most x ” curve, the CDF, $F_X(x)$.

Use it when / how: Walked through using Exam 1, Problem 1 (three separate spots of guaranteed probability, plus a smooth curve in between): **Step 1 — add up what's already accounted for.** The three fixed spots already total $\frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4}$ of all the probability. **Step 2 — the rest has to come from the smooth part.** Every PDF must satisfy $\int f_X(x) dx = 1$, so set the smooth part's area equal to what's left over ($\frac{3}{4}$) and solve — this gives the missing number, $A = 9/64$. **Step 3 — build the curve piece by piece.** $F_X(x) = \int f_X$: add up the smooth part as you go, and every time you pass one of the three fixed spots, jump up by that spot's full amount. **Step 4 — answer the actual question.** Read F_X at the value you need straight off the curve you just built (or subtract two values of F_X).

★ Type 2 — Boil a Two-Variable Problem Down to a Few Numbers ★ [PROFESSOR-FLAGGED]

Seen in: Exam 1, Problems 2 and 3; Exam 2, Problems 1, 4, and 5 (5 of 12 — the most common type by far)

Example: You're given a rule connecting two variables, but the question only asks for averages ($E[X]$), spread ($\text{Var}[X]$), or how the two move together ($\text{Cov}[X, Y]$). You don't need to solve the whole two-variable puzzle — just boil it down to a handful of plain numbers and combine them with simple arithmetic.

Use it when / how: Walked through using Exam 2, Problem 1 (the rule $f(x, y) = 4xy$ on the unit square): **Step 1 — get each variable on its own, if needed.** Add up (or integrate over) the other variable to get one variable's own curve, $f_X(x)$ (this is called the *marginal*). (Not even needed here, since the plain averages below can be read straight off the two-variable rule.) **Step 2 — compute the plain averages.** The average of X ($E[X]$), the average of Y ($E[Y]$), the average of X^2 ($E[X^2]$), and the average of X times Y ($E[XY]$), each found by direct calculation: $E[X] = 2/3$, $E[X^2] = 1/2$, $E[XY] = 4/9$. **Step 3 — combine with simple formulas.** Spread of X ($\text{Var}[X] = E[X^2] - (E[X])^2 = 1/2 - (2/3)^2 = 1/18$). How X and Y move together ($\text{Cov}[X, Y] = E[XY] - E[X]E[Y] = 4/9 - (2/3)^2 = 0$). **Step 4 — combine further if asked.** Spread of $X + Y$ ($\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y] = 1/18 + 1/18 + 2(0) = 1/9$).

Type 3 — Matching a Two-Variable Bell-Curve Formula

Seen in: Exam 2, Problem 2 (1 of 12, but this exact style is one of the professor's personal must-knows — treat it as very likely to reappear)

Example: You're handed a two-variable, bell-curve-shaped formula with an unknown front constant and some unknown “spread” (σ_X, σ_Y) and “how-they-relate” ($\rho_{X,Y}$) numbers hidden inside it. Line it up against the standard bell-curve formula, piece by piece, to read those numbers off.

Use it when / how: Walked through using Exam 2, Problem 2 (the rule $ce^{-(2x^2 - 4xy + 4y^2)}$): **Step 1 — recognize the shape.** A constant times “ e ” to a negative quadratic in x and y — here the quadratic is $2x^2 - 4xy + 4y^2$. **Step 2 — line it up against the standard formula.** The standard two-variable bell curve has known slots for σ_X , σ_Y , and $\rho_{X,Y}$; matching term by term gives $\sigma_X = 1/\sqrt{2}$, $\sigma_Y = 1/2$, $\rho_{X,Y} = 1/\sqrt{2}$. **Step 3 — get the front constant.** Plug those three numbers into the standard formula's constant to get $c = 2/\pi$. **Step 4 — state the conclusion.** Since $\rho_{X,Y}$ isn't zero, the two variables are *not* independent — knowing one tells you something real about the other.

★ Type 4 — Turning One Random Variable Into Another ★ [PROFESSOR-FLAGGED]

Seen in: Exam 1, Problem 7; Exam 2, Problem 3 (2 of 12 — consistently the hardest, last problem on each exam)

Example: You're given a rule for building a new variable $W = g(X)$ out of an old one (e.g. squaring it, or running it through some circuit), and asked for the *whole probability curve* of the new variable, $f_W(w)$ — not just its average.

Use it when / how: Walked through using Exam 2, Problem 3 (X is the current, evenly spread between -2 and 2 ; $Y = 9X^2$ is the power): **Step 1** — figure out what values the new variable can take. Since X is between -2 and 2 , $Y = 9X^2$ can only land between 0 and 36 . **Step 2** — rewrite “ Y at most y ” ($F_Y(y)$) as a statement about the old variable. “ $Y \leq y$ ” means “ X is between $-\sqrt{y/9}$ and $\sqrt{y/9}$ ”; since X is evenly spread, that chance is just (length of that stretch)/(total length), giving $F_Y(y) = \sqrt{y}/6$. **Step 3** — watch the edges. If those square-root bounds ever stick out past where the old variable is actually allowed to be, stop them at its edge instead (didn't happen here, but often does — check every time). **Step 4** — take the derivative. $f_Y(y) = dF_Y/dy$ turns $\sqrt{y}/6$ into the final answer, $1/(12\sqrt{y})$. **Step 5** — if the problem then caps the new variable at some value (like a meter that can't read above 16), the curve f stays the same below the cap, and one extra pile of leftover probability (size $1 - 4/6 = 1/3$ here) sits right at the cap.

Type 5 — Bell-Curve Probabilities Way Out in the Tail

Seen in: Exam 1, Problem 5 (1 of 12)

Example: Ordinary bell-curve (Gaussian) random variable, but the question asks about something rare or far from the average — like an error rate.

Use it when / how: Walked through using Exam 1, Problem 5 (bit-error rate vs. signal strength): **Step 1** — rescale to “how many spreads from the average” ($z = (x - \mu)/\sigma$). Subtract the average μ , divide by the spread σ . **Step 2** — use the far-tail table (Q), not the middle-of-the-curve one (Φ), whenever the question is about something rare (“chance of error”). With signal strength 4 , this gives $Q(2) \approx 0.0228$. **Step 3** — if the question runs backward (“how strong must the signal be so the error rate is under one in a million?”), look up the Q -table *backward* to find the cutoff point z first (4.75), then solve for the original quantity (signal strength must be at least 22.56).

Type 6 — Using the Far Edges of a Two-Variable Curve to Find a Constant

Seen in: Exam 1, Problem 6 (extra credit) (1 of 12)

Example: You're given a two-variable “chance both are at most these values” formula ($F_{X,Y}(x,y)$) with an unknown constant, and told that way out at the far edges, the total chance must reach exactly 1 : $F_{X,Y}(\infty, \infty) = 1$.

Use it when / how: Walked through using Exam 1, Problem 6: **Step 1** — push both variables out to plus-infinity in the given formula and simplify (here, using that an arctangent approaches a known fixed value far out). **Step 2** — set that equal to 1 and solve for the constant, giving $a = 1/\pi^2$. **Step 3** — sanity-check the other edges: push the variables to minus-infinity instead and confirm you get 0 , as any such formula must ($F_{X,Y}(-\infty, y) = 0$).

Chapter 13: Three Problem Types (from the Homework Set)

Type 7 — Checking Whether a Formula Could Really Describe a Self-Correlation

Seen in: Homework 13.9.1 (1 of 6, but tests the professor's personally-flagged theorem — treat it as very likely to reappear)

Example: You're handed a candidate formula and asked whether it's even *allowed* to be an autocorrelation, written $R_X(\tau)$ or $R(\tau)$ — a measure of how a signal lines up with a delayed copy of itself, τ seconds later.

Use it when / how: Walked through using Homework 13.9.1 (four candidate formulas for $R(\tau)$): **Step 1** — check that flipping the sign of the delay doesn't change the answer: $R(\tau)$ must equal $R(-\tau)$. One candidate here failed immediately: shifting a spike to $\tau = +10$ instead of $\tau = -10$ gave a different formula. **Step 2** — check that $R(0)$ (“no delay at all”) is the single biggest value the formula ever reaches: $R(0) \geq |R(\tau)|$. **Step 3** — check that, far away, $R(\tau)$ settles down to something zero or positive — never negative ($\lim_{|\tau| \rightarrow \infty} R(\tau) \geq 0$). One candidate here settled down to -10 , which is impossible, so it was ruled out too.

★ Type 8 — Splitting an Average When Two Things Don't Affect Each Other ★ [PROFESSOR-FLAGGED]

Seen in: Homework 13.9.5, 13.9.9, and 13.10.1 (3 of 6 — the single most common Chapter 13 skill)

Example: You need the average of a product of two things ($E[AB]$), and those two things have no effect on each other at all (they're *independent*) — so you're allowed to average each one separately and just multiply the results: $E[AB] = E[A]E[B]$.

Use it when / how: Walked through using Homework 13.10.1 (two completely separate signals $X(t), Y(t)$ multiplied together, $W(t) = X(t)Y(t)$): **Step 1** — confirm exactly what doesn't affect what. Here, the two whole signals are unrelated to each other. **Step 2** — split the average of the product into a product of averages. The autocorrelation of W , $R_W(\tau) = E[W(t)W(t+\tau)] = E[X(t)X(t+\tau)] \cdot E[Y(t)Y(t+\tau)] = R_X(\tau)R_Y(\tau)$ — signal 1's own self-correlation times signal 2's own self-correlation. **Step 3** — watch for a special case. Sometimes “no delay” ($\tau = 0$, or $k = 0$ for a sequence) behaves differently from “some delay,” or a spinning random angle averages away to exactly zero — handle that piece on its own when it comes up. **Step 4** — put the pieces back together to get the final formula.

Type 9 — Following a Time-Shift Through the Formula

Seen in: Homework 13.10.3 and 13.11.2 (2 of 6)

Example: A new signal is built by delaying, speeding up/slowing down, or combining samples of an old one. Track exactly which time-gap ends up mattering inside $R_X(\cdot)$, or confirm the new signal is still bell-curve-shaped (Gaussian) if the old one was.

Use it when / how: Walked through using Homework 13.10.3 (a plain delayed copy of a signal, $Y(t) = X(t - t_0)$): **Step 1** — write down the two specific time points being compared (for $R_Y(\tau) = E[Y(t)Y(t+\tau)]$, that's $t - t_0$ and $t + \tau - t_0$). **Step 2** — subtract them to get the time-gap that actually matters: here it's just τ again. A pure delay applied to *both* copies leaves this gap — and so $R_X(\cdot)$ — completely unchanged: $R_Y(\tau) = R_X(\tau)$. Comparing the original to a delayed copy instead ($R_{XY}(\tau) = E[X(t)Y(t+\tau)]$) shifts the gap by however long the delay is: $R_{XY}(\tau) = R_X(\tau - t_0)$. (A related question, Homework 13.11.2, asks about combining or re-timing a Gaussian signal instead: doubling it, subtracting two of its samples, or sampling it at different times never breaks the “still Gaussian” property — that part is automatic.)

If a Final-Exam Question Doesn't Match Any of These Nine Exactly

It's almost certainly a small variation on one of them. Ask yourself: is this a one-variable-curve question (Type 1)? A two-variable-averages question (Type 2 — the most likely by far)? A bell-curve-formula-matching question (Type 3)? A build-a-new-variable question (Type 4)? A far-tail bell-curve question (Type 5)? A find-the-constant-from-the-edges question (Type 6)? For Chapter 13: a check-the-formula question (Type 7)? A split-the-average question (Type 8 — the most likely of the three)? Or a follow-the-time-shift question (Type 9)? Once you know which one it is, the steps above tell you exactly what to do.

Companion document: *Final_Exam_Guide.pdf* — full proofs, complete worked examples, and every Exam 1 / Exam 2 problem solved step by step.